Intro to AI in Medicine

Lec 17: When to use AI/Final Lecture

Last Time

- Interactive case study
- Practiced talking to people with different backgrounds, who were experts in different areas and synthesizing recommendations
- Learned that stakeholders have variable motivations
- Any thoughts 2 days later?

This time

- Last lecture
- Discussing intuition on AI and when to use it – and when not to

Final Project

- Due Sunday 11:59 PM to me over email
- Will stay today to answer questions
- No other office hours but will be reachable via email, text, dm
- If you will have <15 classes attended by next Monday, passing is higher for you than 70%

Question

- What are some ways this semester we’ve seen AI fail to achieve it’s intended goal?

AI Failure Modes

- Low awareness of clinical consequences
- Data that does not reflect reality
- Fitting to noise, not signal
- Imprecision in a well-defined workflow (use what instead?)
- Goals of AI and humans misaligned
- Resource intensive

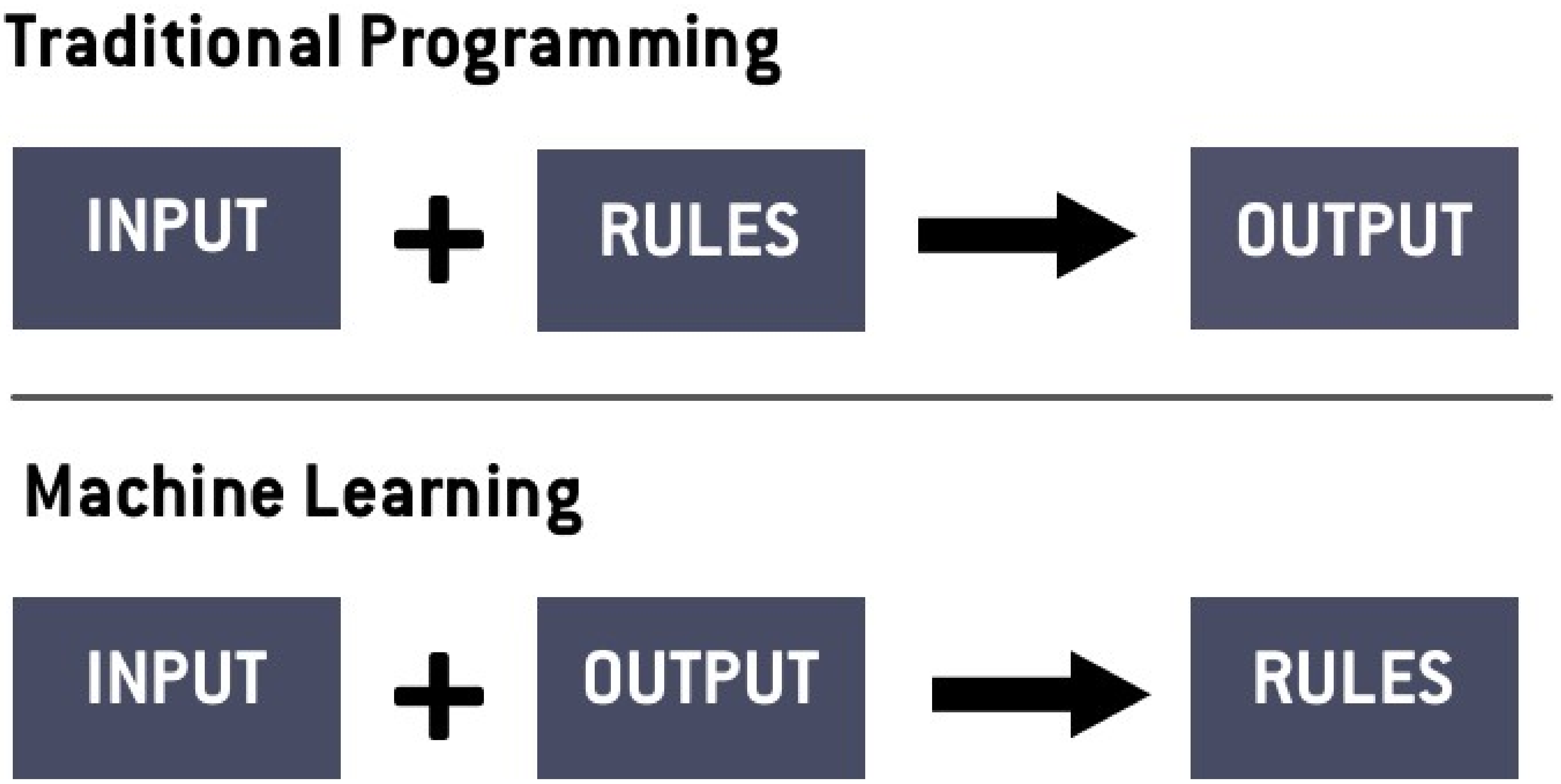
General Problem Statement

Technical Problem Statement

Data	Loss
Hallucinations Bias Underfit Drift	Alignment
Network	**Training**
Overfit	PHI protection Financial/Environmental costs

When is AI Feasible - Intuition

- Short answer – if you can solve data and loss, the rest is easy
- Questions to ask yourself:
  - Can we come up with a mathematical definition of success?
  - Are there tons of examples of what success and failure are?
  - Can we do these quickly?
- This often happens when tasks are *digitizable*
- When is an algorithm better:
  - You need perfection
  - You can create a “recipe” of how to get from start to finish


Traditional Programming
INPUT
+
RULES
OUTPUT
Machine Learning
INPUT
+
OUTPUT
RULES

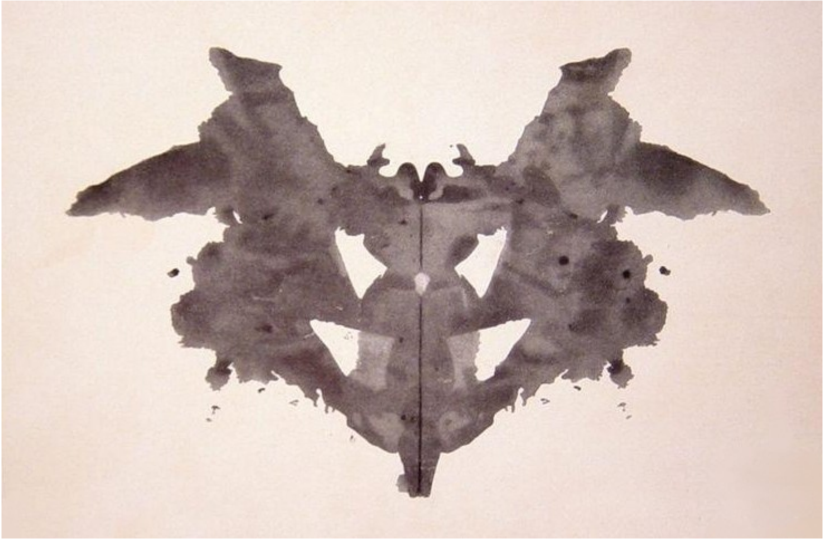
In More Human Terms…

- If a human had:
  - infinite attention span
  - Infinite time
- And their job was:
  - Give things a score
  - Place into one of many classes

- Then an AI can probably be trained to do it.

In More Human Terms…

- Examples:
    - Rorschach blots
    - Pictures of trees



The Future of AI – Who will be replaced?

- Low skill, white collar work
- High skill, white collar work
- Low skill, blue collar work
- High skill, blue collar work

- AI rate limiters are data and digitization (corollary to loss)
- Typically, data moves faster than digitization

Table 3: Top 40 occupations with highest AI applicability score.

Job Title (Abbrv.)	Coverage	Cmpltn.	Scope	Score	Employment
Interpreters and Translators	0.98	0.88	0.57	0.49	51,560
Historians	0.91	0.85	0.56	0.48	3,040
Passenger Attendants	0.80	0.88	0.62	0.47	20,190
Sales Representatives of Services	0.84	0.90	0.57	0.46	1,142,020
Writers and Authors	0.85	0.84	0.60	0.45	49,450
Customer Service Representatives	0.72	0.90	0.59	0.44	2,858,710
CNC Tool Programmers	0.90	0.87	0.53	0.44	28,030
Telephone Operators	0.80	0.86	0.57	0.42	4,600
Ticket Agents and Travel Clerks	0.71	0.90	0.56	0.41	119,270
Broadcast Announcers and Radio DJs	0.74	0.84	0.60	0.41	25,070
Brokerage Clerks	0.74	0.89	0.57	0.41	48,060
Farm and Home Management Educators	0.77	0.91	0.55	0.41	8,110
Telemarketers	0.66	0.89	0.60	0.40	81,580
Concierges	0.70	0.88	0.56	0.40	41,020
Political Scientists	0.77	0.87	0.53	0.39	5,580
News Analysts, Reporters, Journalists	0.81	0.81	0.56	0.39	45,020
Mathematicians	0.91	0.74	0.54	0.39	2,220
Technical Writers	0.83	0.82	0.54	0.38	47,970
Proofreaders and Copy Markers	0.91	0.86	0.49	0.38	5,490
Hosts and Hostesses	0.60	0.90	0.57	0.37	425,020
Editors	0.78	0.82	0.54	0.37	95,700
Business Teachers, Postsecondary	0.70	0.90	0.52	0.37	82,980
Public Relations Specialists	0.63	0.90	0.60	0.36	275,550
Demonstrators and Product Promoters	0.64	0.88	0.53	0.36	50,790
Advertising Sales Agents	0.66	0.90	0.53	0.36	108,100
New Accounts Clerks	0.72	0.87	0.51	0.36	41,180
Statistical Assistants	0.85	0.84	0.49	0.36	7,200
Counter and Rental Clerks	0.62	0.90	0.52	0.36	390,300
Data Scientists	0.77	0.86	0.51	0.36	192,710
Personal Financial Advisors	0.69	0.88	0.52	0.35	272,190
Archivists	0.66	0.88	0.49	0.35	7,150
Economics Teachers, Postsecondary	0.68	0.90	0.51	0.35	12,210
Web Developers	0.73	0.86	0.51	0.35	85,350
Management Analysts	0.68	0.90	0.54	0.35	838,140
Geographers	0.77	0.83	0.48	0.35	1,460
Models	0.64	0.89	0.53	0.35	3,090
Market Research Analysts	0.71	0.90	0.52	0.35	846,370
Public Safety Telecommunicators	0.66	0.88	0.53	0.35	97,820
Switchboard Operators	0.68	0.86	0.52	0.35	43,830
Library Science Teachers, Postsecondary	0.65	0.90	0.51	0.34	4,220

https://arxiv.org/pdf/2507.07935

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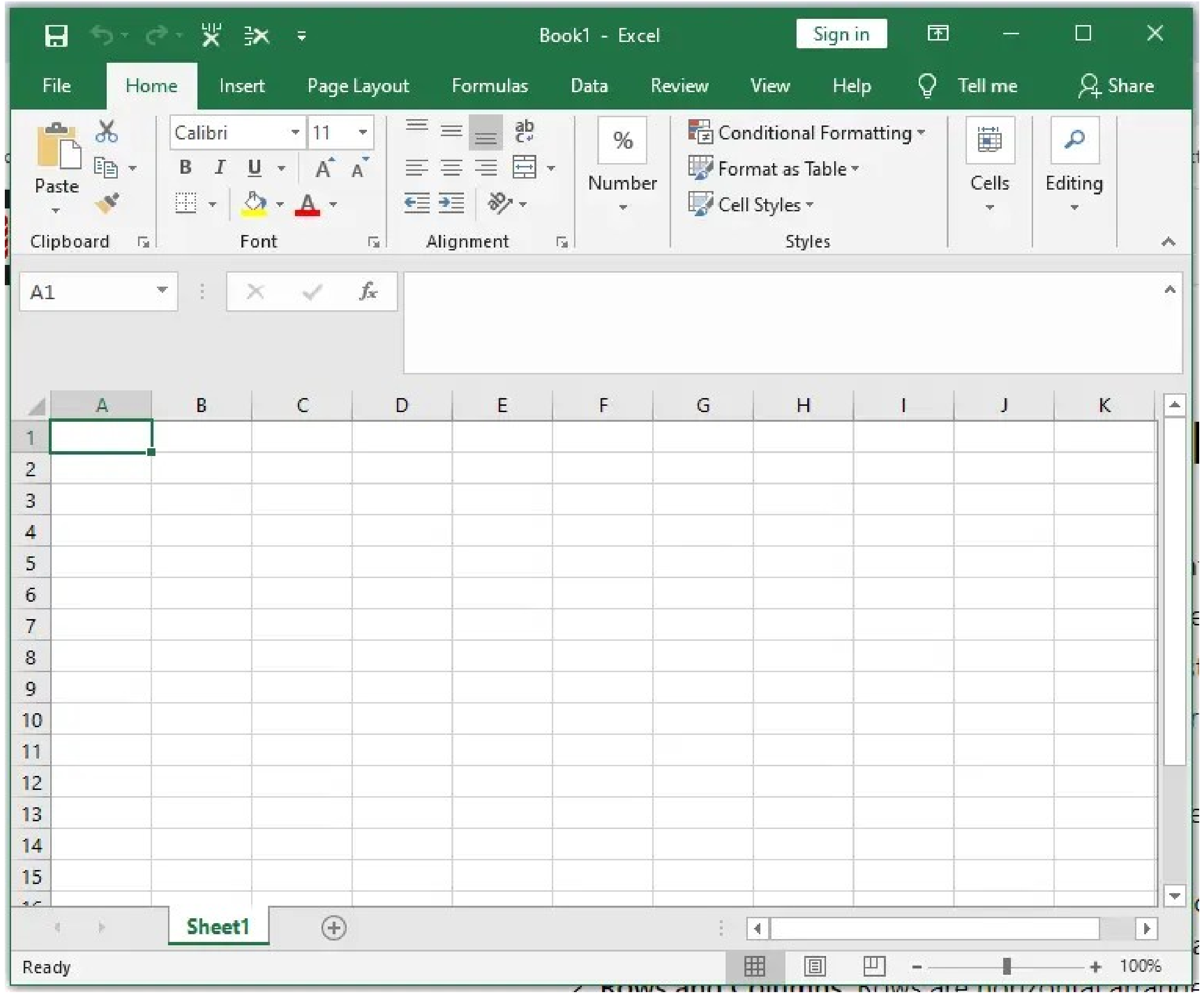
Supercharge Your Workforce



TOP 10 JOBS AT RISK OF AI DISPLACEMENT

1. Data Entry Clerks
2. Telemarketers
3. Customer Service Representatives
4. Retail Cashiers
5. Proofreaders and Copy Editors
6. Paralegals and Legal Assistants
7. Bookkeepers
8. Fast Food and Restaurant Workers
9. Warehouse Workers
10. Market Research Analysts

Directions for AI

- Network Architectures
- Domain specific AI applications
- AI Domain transfer
- Alignment
- Human AI interaction

- I hope you all will contribute to these in some way!

Course Feedback

- What lectures were most memorable? (not including last one)
- What should have been covered in more depth? What can be skipped?
- Biggest takeaway from the course?
- What kind of people would benefit from this class? (career, personal interest)
- What kind of assignments make sense if this class is a regular college course?
- Any other comments?

Thanks!